

L1.5 Differentiation Formulae

Work in your group. Check your answers against the Quarto tonight.

The five rules

1. Differentiate term by term.

(a) $y = x^4 + 2x^2 + 3x$

(b) $f(x) = 30x^3 - 6x^2 + 11$

Stop. Can each of these be written as a single power of x before you differentiate anything?

2. Rewrite each as a power of x first, then differentiate.

(a) $\sqrt{x}$ (b) $\frac{1}{x^3}$ (c) $\sqrt[3]{x^2}$ (d) $\frac{5}{x^2}$

3. $y = 20\sqrt{x}$. Find $\left. \frac{dy}{dx} \right|_{x=2}$.

Product and quotient rules

4. $f(x) = (x^2 - 1)(x + 4)$. Differentiate it **two ways**:

- (a) with the product rule
- (b) by multiplying it out first, then differentiating term by term

Do they agree? **What would you have got by multiplying the two derivatives together?**

Stop. Which rule does this need — and why not the other one?

5. Differentiate.

- (a) $f(x) = (3x^2 + 1)(x^3 - 2x)$
- (b) $y = \sqrt{x}(x^2 + 3)$

6. Differentiate. **One of these does not need the quotient rule at all — say which, and why.**

(a) $y = \frac{x^2 + x - 2}{x + 4}$

(b) $y = \frac{x^2 + 3}{x}$

7. Name the rule each one needs, and say why not the other one. Do not compute anything.

$$x^3\sqrt{x}$$

$$\frac{x^5 + x}{x^2}$$

$$(2x + 1)(x^3 - x)$$

$$\frac{2x + 1}{x^3 - x}$$

Higher derivatives and horizontal tangents

8. Find f'' .

(a) $f(x) = x^4 - 3x^2 + 1$

(b) $f(x) = x^3 - 2x^2 - 3x - 1$ (*keep this one — you need it again at the boards*)

9. $y = x^3 - x$. Find every horizontal tangent.

10. Find the equation of the tangent line to $f(x) = x^3 - 3x + 4$ at $x = 2$.

Last lesson this took four lines of limits. How many does it take now?

11. **BOARDS** $f(x) = x^3 - 2x^2 - 3x - 1$, from item 8(b).

(a) Graph f . Mark every inflection point (IP) and every local max or min (M).

(b) Graph f' underneath it. **What happened to the IPs?**

(c) Graph f'' underneath that. **What happened to the Ms?**

(d) Complete the sentence.

IPs become _____ on the first derivative,

which become _____ on the second.

12. **BOARDS** Now do the same for $g(x) = x^3 - 3x - 1$. **What is different, and what is the same?**